

Ex 1.

Le tableau en dessous indique la distribution de la population selon les salaires :

$x_i \cdot 10^3$	$[4; 6[$	$[6; 8[$	$[8; 10[$	$[10; 12[$	$[12; 14[$
n_i	5	8	12	10	8
c_i	5	7	9	11	13
f_i	0,11	0,19	0,28	0,23	0,19
F_i	0,11	0,3	0,58	0,71	1
$n_i c_i$	25	56	108	110	104
$\frac{n_i c_i}{\sum n_i c_i}$	0,06	0,14	0,27	0,28	0,26
Q_i	0,06	0,2	0,47	0,74	1

$N = 43$

$\sum n_i c_i = 403$

1) Population = 43 salarier

2) Caractere = Salaire

Nature = Quantitatif continue

3)

Mediane :

Classe mediane : $\frac{43}{2} = 21,5 \rightarrow [8, 10[$

$M_e = 8 + 2 \left(\frac{0,5 - 0,3}{0,58 - 0,3} \right) = 8 + 2 \left(\frac{5}{7} \right)$

$\Rightarrow M$

$M_e = 942,86$

Mediale :

$$\text{Classe mediale} : \frac{403}{2} = 201,5 \rightarrow [10, 12[$$

$$M_b = 10 + 2 \left(\frac{0,5 - 0,47}{0,74 - 0,47} \right) = 10 + 2 \left(\frac{1}{9} \right) =$$

$$M_b = 1022,22$$

5)

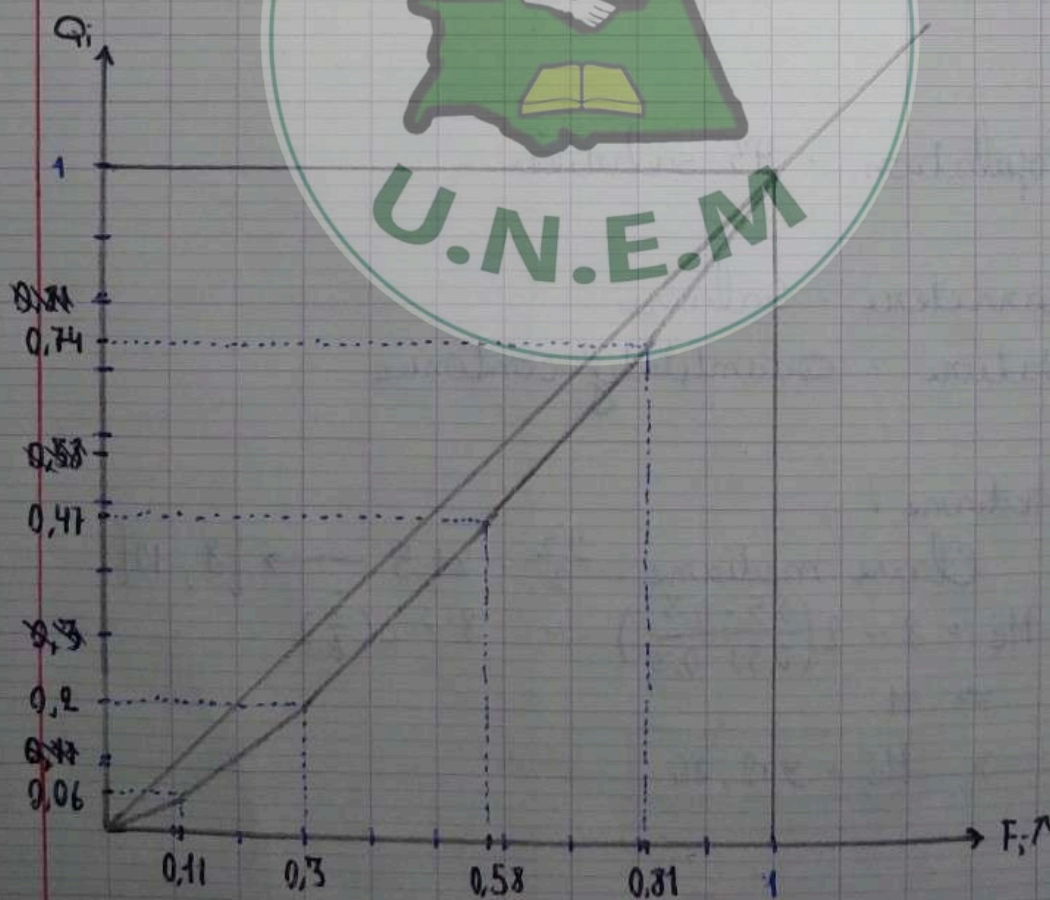
Indice de concentration :

$$I_c = \frac{M_b - M_e}{w} = \frac{1022,22 - 942,86}{1400 - 400}$$

$$\Rightarrow I_c = 7,93\%$$

6)

$$I_G = 1 - \sum_{i=1}^n (q_i + q_{i+1})(F_i - F_{i-1})$$



$$I_G = 1 - \sum_{i=1}^n f_i (a_i + a_{i-1})$$

$$\begin{aligned} \Rightarrow I_G &= 1 - (0,06 \times 0,11 + 0,26 \times 0,19 + 0,67 \times 0,28 + 1,21 \times 0,23 + 1,74 \times 0,19) \\ &= 1 - 0,60203 \\ \Rightarrow I_G &= 0,39797 \end{aligned}$$

Serie 1 $\rightarrow$ Stat $\rightarrow$ TD $\rightarrow$ FC/L1 $\rightarrow$ S2 $\rightarrow$ 2025

Ex 1

x: Vitesse km/h	40	50	60	70	80	90	100	110	120
y: distance m	8	12	18	24	32	40	48	58	72
$x_i y_i$	320	600	1080	1680	2560	3600	4800	6380	8640
x_i^2	1600	2500	3600	4900	6400	8100	10000	12100	14400
y_i^2	64	144	324	576	1024	1600	2304	3364	5184

29.660

63.600

14.584

1-

①

Methode de points extremes:

$y = ax + b$ passant par $(x_{\min}, y_{\min})$ et $(x_{\max}, y_{\max})$

$$a = \frac{y_{\max} - y_{\min}}{x_{\max} - x_{\min}} = \frac{72 - 8}{120 - 40} = \frac{64}{80} = 0,8 \Rightarrow a = 0,8$$

$$y_{\min} = 0,8 \times x_{\min} + b \Rightarrow b = y_{\min} - 0,8 x_{\min} = 8 - 0,8 \times 40 = -24$$

$$\Rightarrow \boxed{y = 0,8x - 24}$$

② Methode de Mayer :

$y = ax + b$ passe par $(\bar{x}_{G_1}, \bar{y}_{G_1})$ et $(\bar{x}_{G_2}, \bar{y}_{G_2})$

$$\bar{x}_{G_1} = (40 + 50 + 60 + 70 + 80) / 5 = 60 \Rightarrow \bar{x}_{G_1} = 60$$

$$\bar{y}_{G_1} = (8 + 12 + 18 + 24 + 32) / 5 = 18,8 \Rightarrow \bar{y}_{G_1} = 18,8$$

$$\bar{x}_{G_2} = (90 + 100 + 110 + 120) / 4 = 105 \Rightarrow \bar{x}_{G_2} = 105$$

$$\bar{y}_{G_2} = (40 + 48 + 58 + 72) / 4 = 54,5 \Rightarrow \bar{y}_{G_2} = 54,5$$

$$a = \frac{\bar{y}_{G_2} - \bar{y}_{G_1}}{\bar{x}_{G_2} - \bar{x}_{G_1}} = \frac{54,5 - 18,8}{105 - 60} = \frac{35,7}{45}$$

$$\Rightarrow a = 0,79$$

$$b = \bar{y}_{G_1} - 0,79 \times \bar{x}_{G_1} = 18,8 - 0,79 \times 60$$

$$\Rightarrow b = -28,6$$

$$y = 0,79x - 28,6$$

③ Drote MCO

$$a = \frac{\text{Cov}(x, y)}{V(x)} ; b = \bar{y} - a\bar{x}$$

$$\text{Cov}(x, y) = \frac{\sum x_i y_i}{N} - \bar{x} \bar{y} \quad V(x) = \frac{\sum x_i^2}{N} - \bar{x}^2$$

$$\Rightarrow \text{Cov}(x, y) = \frac{23660}{9} - \frac{720}{9} \times \frac{312}{9} = 522,22$$

$$\Rightarrow V(x) = \frac{63600}{9} - \left(\frac{720}{9}\right)^2 = 666,67$$

$$\Rightarrow a = \frac{522,22}{666,67} = 0,78 \Rightarrow b = \frac{312}{9} - 0,78 \times \frac{720}{9} = -27,73$$

$$\Rightarrow y = 0,78x - 27,73$$

$$r = \frac{\text{Cov}(x, y)}{s(x) \times s(y)}$$

$$s(x) = \sqrt{v(x)} = \sqrt{666,67} = 25,82$$

$$s(y) = \sqrt{v(y)} = \sqrt{\frac{\sum y_i^2}{N} - \bar{y}^2} = \sqrt{\frac{84,6^2}{9} - \left(\frac{312}{9}\right)^2}$$

$$= \sqrt{\frac{14584}{9} - \left(\frac{312}{9}\right)^2} = 20,46$$

$$\text{Cov}(x, y) = 522,99$$

$$\Rightarrow r = \frac{522,99}{25,82 \times 20,46} = 0,988$$

$\Rightarrow r$ est une très forte corrélation linéaire de sens (+)
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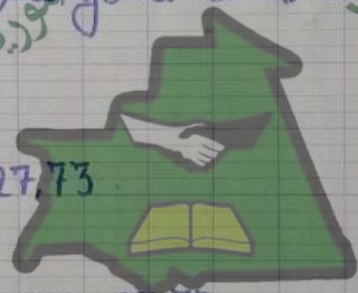
3)

$$Y = 0,78x - 27,73$$

$$x = 130 \text{ km/h}$$

$$\Rightarrow Y = 0,78 \times 130 - 27,73$$

$$Y = 73,67 \text{ m}$$

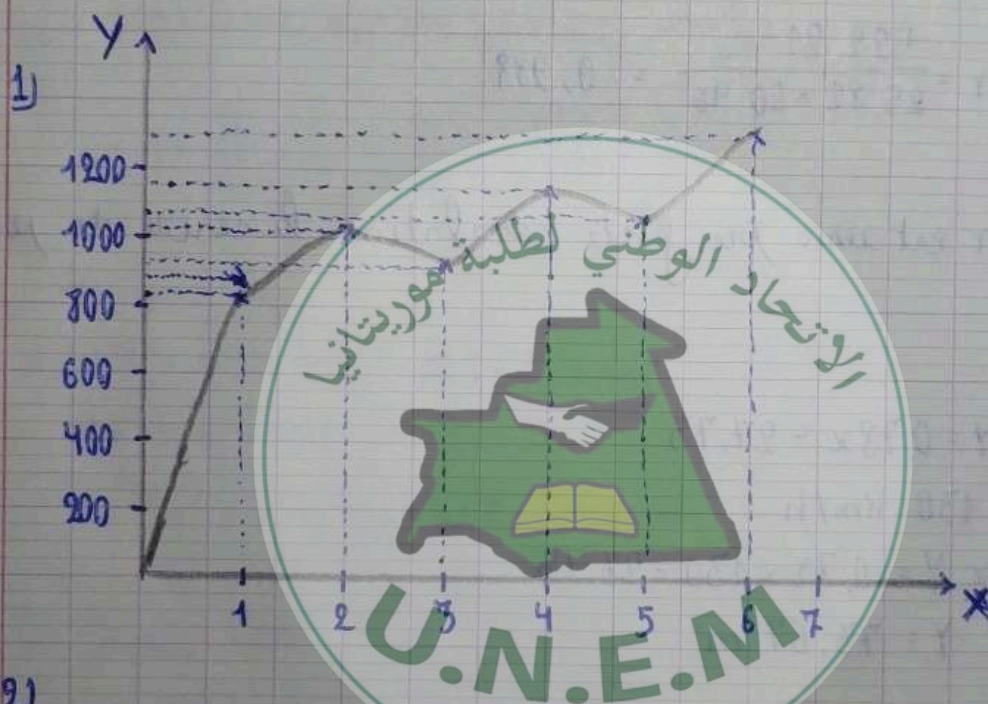


U.N.E.M

2x2

x	1	2	3	4	5	6
y	842	1095	922	1154	1058	1298
xy	842	2190	2766	4616	5290	7788
x ²	1	4	9	16	25	36
y ²	708.964	1.199.025	850.084	1.331.716	1.119.364	1.684.804

6893957



2)

Etant donné que les points se regroupent autour d'une droite, le choix de l'ajustement linéaire est justifié.

$$r = \frac{\text{Cov}(x, y)}{s(x) \cdot s(y)}$$

$$\text{Cov}(x, y) = \frac{\sum x_i y_i}{N} - \bar{x} \bar{y} = \frac{23492}{6} - \frac{21}{6} \times \frac{6369}{6} = 200,08$$

$$v(x) = \frac{\sum x_i^2}{N} - \bar{x}^2 = \frac{91}{6} - \left(\frac{21}{6}\right)^2 = 2,92, \quad s(x) = \sqrt{2,92} = 1,7$$



$$V(y) = \frac{\sum y_i^2}{N} - \bar{y}^2 = \frac{6893957}{6} - \left(\frac{6369}{6}\right)^2 = 22210,58$$

$$S(y) = \sqrt{22210,58} = 149,03$$

$$\Rightarrow r = \frac{900,08}{1,7 \times 149,03} = 0,79$$

$\Rightarrow$ Le choix de l'ajustement linéaire est confirmé par le résultat analytique obtenu.

3)

* Par MCO

$$a = \frac{\text{Cov}(x, y)}{V(x)} = \frac{900,08}{2,9} = 68,75$$

Comme $(\bar{x}, \bar{y}) = (7, 6369/6)$, alors $b = \bar{y} - a\bar{x}$

$$\Rightarrow b = \frac{6369}{6} - 68,75 \times \frac{21}{6} = 820,87$$

$$\Rightarrow y = 68,75x + 820,87$$

4)

Si $x = 7$

$$\Rightarrow y = 68,75(7) + 820,87 \Rightarrow y = 1302,12$$

Si $x = 8$

$$\Rightarrow y = 68,75(8) + 820,87 \Rightarrow y = 1370,87$$

Ex 4:

a:

$$D: Y = X + 30$$

$$\rightarrow Y = aX + b$$

$$D': X = \frac{1}{4}Y + 60$$

$$\rightarrow X = a'Y + b'$$

1)

Coefficient de corrélation linéaire :

$$r = \sqrt{a \times a'}$$

$$= \sqrt{1 \times \frac{1}{4}} \Rightarrow r = 0,5$$

2)

Calcul de $\bar{x}$ et $\bar{y}$

$$\begin{cases} b = \bar{y} - a\bar{x} \\ b' = \bar{x} - a'\bar{y} \end{cases} \Rightarrow \begin{cases} \bar{y} = b + a\bar{x} \\ \bar{x} = b' + a'\bar{y} \end{cases}$$

$$\Rightarrow \bar{y} = b + a(b' + a'\bar{y})$$

$$= b + ab' + aa'\bar{y}$$

$$\Rightarrow \bar{y} = 30 + 1 \times 60 + 1 \times \frac{1}{4}\bar{y}$$

$$\bar{y} = 90 + \frac{1}{4}\bar{y} \Rightarrow \frac{3}{4}\bar{y} = 90$$

$$\Rightarrow \bar{y} = 120$$

$$\bar{x} = b' + a'\bar{y} = 60 + \frac{1}{4}(120) = 60 + 30$$

$$\Rightarrow \bar{x} = 90$$

$$\begin{cases} \bar{x} = 90 \end{cases}$$

$$\begin{cases} \bar{y} = 120 \end{cases}$$

3)

Calcul de $\text{Cov}(x, y)$ et $v(x)$

On a: $v(y) = 40$

$$a' = \frac{\text{Cov}(x, y)}{v(y)} \Rightarrow \text{Cov}(x, y) = a' v(y)$$

$$\text{Cov}(x, y) = \frac{1}{4} \times 40 \Rightarrow \boxed{\text{Cov}(x, y) = 10}$$

$$a = \frac{\text{Cov}(x, y)}{v(x)} \Rightarrow v(x) = \frac{\text{Cov}(x, y)}{a}$$

$$v(x) = \frac{10}{1} \Rightarrow \boxed{v(x) = 10}$$

Ex 5 :

Norm	A	B	C	D	E	F	G	H	I	J
Stat $\rightarrow x$	10	8	3	6	2	4	7	1	5	9
Math $\rightarrow y$	9	7	4	5	1	6	10	2	3	8
r_x	10	8	3	6	2	4	7	1	5	9
r_y	9	7	4	5	1	6	10	2	3	8
$r_x - r_y = d_i$	1	1	-1	1	1	-2	-3	-1	2	1
d_i^2	1	1	1	1	1	4	9	1	4	1

= 24

$$r_{sp} = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 24}{10(100 - 1)} = 1 - 0,145$$

$$\Rightarrow r_{sp} = 0,855$$

Il s'agit d'une très forte corrélation de sens (+)

Methode ②

$$r_{sp} = \frac{Cov(r_x, r_y)}{s(r_x) \cdot s(r_y)}$$

$$Cov(r_x, r_y) = \frac{\sum r_x r_y}{N} - \bar{r}_x \cdot \bar{r}_y$$

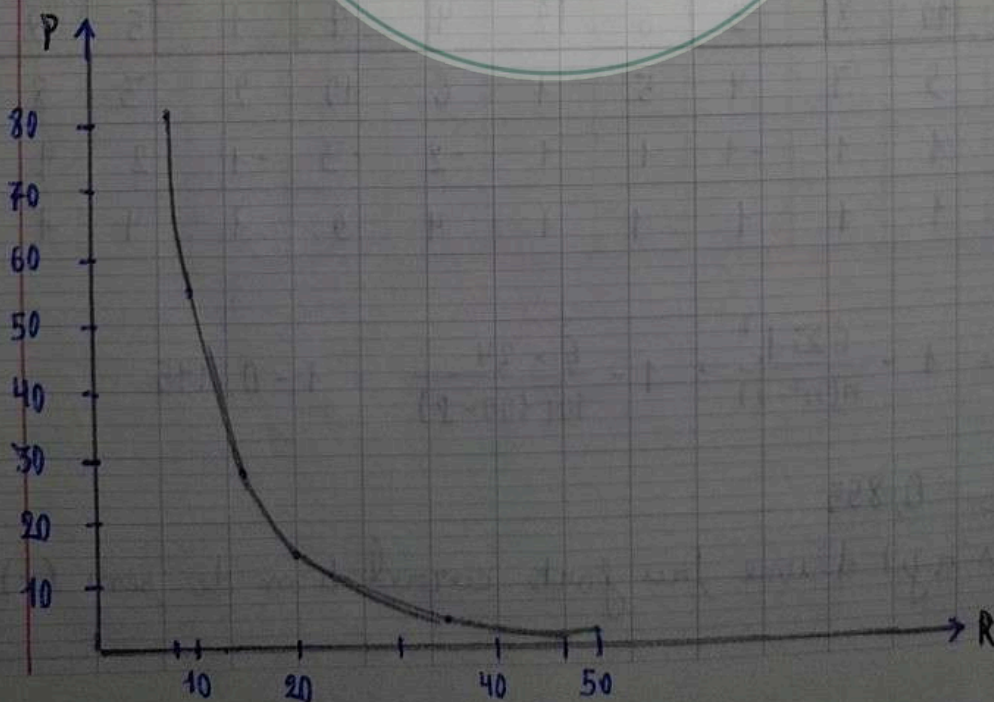
$$s(r_x) = \sqrt{v(r_x)} = \sqrt{\frac{\sum r_x^2}{N} - \bar{r}_x^2}$$

$$s(r_y) = \sqrt{v(r_y)} = \sqrt{\frac{\sum r_y^2}{N} - \bar{r}_y^2}$$

مثال 9:

		8	10	15	20	35	50	Total
R		8	10	15	20	35	50	
P		80,3	55,8	28,1	15,2	5,1	2,5	
ln(R)		2,08	2,3	2,7	2,99	3,55	3,91	17,53
ln(P)		4,38	4,02	3,33	2,72	1,63	0,91	16,92
ln(R) × ln(P)		9,11	9,246	8,99	8,13	5,78	3,56	44,816
ln ² (R)		4,326	5,29	7,29	8,94	12,6	15,29	53,736
ln ² (P)		19,18	16,16	11,09	7,4	2,657	0,83	57,317

1)



Un nuage linéaire ne suggère pas un enregistrement linéaire car les points ne se regroupent pas autour d'une droite

2)

$$r = \frac{\text{Cov}(\ln(R_i); \ln(P_i))}{\delta(\ln(R_i)) \times \delta(\ln(P_i))} \quad \begin{cases} x_i = \ln(R_i) \\ y_i = \ln(P_i) \end{cases}$$

$$\text{Cov}(\ln(R_i); \ln(P_i)) = \frac{\sum \ln(R_i) \ln(P_i)}{N} - \overline{\ln(R_i)} \times \overline{\ln(P_i)}$$

$$\text{Cov}(\ln(R_i); \ln(P_i)) = \frac{44,846}{6} - \left(\frac{17,53}{6}\right) \left(\frac{16,99}{6}\right) = -0,8$$

$$V(\ln(R_i)) = \frac{\sum \ln^2(R_i)}{N} - \overline{\ln(R_i)}^2 = \frac{53,736}{6} - \left(\frac{17,53}{6}\right)^2 = 0,42$$

$$V(\ln(P_i)) = \frac{\sum \ln^2(P_i)}{N} - \overline{\ln(P_i)}^2 = \frac{57,317}{6} - \left(\frac{16,99}{6}\right)^2 = 1,53$$

$$r = \frac{-0,8}{\sqrt{0,42} \times \sqrt{1,53}} = -0,99$$

$\Rightarrow r = -0,99 \rightarrow$ Très forte corrélation de sens (-)

3)

$$a = \frac{\text{Cov}(\ln(R_i); \ln(P_i))}{V(\ln(R_i))} = \frac{-0,8}{0,42} = -1,9$$

$$b = \overline{\ln(P_i)} - a(\overline{\ln(R_i)}) = \frac{16,99}{6} - (-1,9) \left(\frac{17,53}{6}\right) = 8,38$$

$$Y = -1,9X + 8,38$$